

REMOTE CONTROLLED MP3 PLAYER With FM Radio

SANI THEO

Have you ever considered making an audio system at home? Building your own

audio system can not only be fun and exciting, it can give you the same performance, if not better, at a

quired for the project before starting its assembly. These are listed in the two boxes for your convenience and ready reference. Significance of the parts is explained below.

Bluetooth MP3 player with FM receiver module. This is the main part of the project. The compact MP3 player module with FM receiver, shown in Fig. 2, is readily available in the market and through many e-commerce portals. It can be installed in your vehicle, desktop PC audio system, home theatre system, or any other audio amplifier.

IR remote control. The IR remote control normally comes along with the Bluetooth MP3 player module. But you can also use some other IR remote to control this MP3 audio player. The author successfully tried three different IR remote models, which are shown in Fig. 3.

Audio amplifier. You can use any audio (power) amplifier having



Fig. 1: Author's prototype



Fig. 2: Bluetooth MP3 player with FM receiver module

much lesser cost. This inexpensive do-it-yourself (DIY) remote-controlled MP3 player cum FM receiver with Bluetooth can be made, including its cardboard box, with easily available components and tools.

The author's prototype for the compact audio system is shown in Fig. 1. It has USB, AUX, and MicroSD/TF (TransFlash) card ports, making it very convenient to interface with external audio systems and storage devices.

Collect all the parts and tools re-



Fig. 3: Common IR remote controls

PARTS LIST

- Bluetooth MP3 player with FM receiver module
- IR remote control
- Audio amplifier
- 5V DC power supply or USB
- Single-strand wire as FM antenna
- 3-wire header cable for audio connection
- 2-wire header cable for power supply
- Type-A USB cable
- 3.5mm stereo audio jack with cable
- Cardboard for making cabinet

TOOLS REQUIRED

- Knife or cardboard cutter
- Scale
- Glue gun
- Scissors
- Soldering iron (25-watt)
- Pencil/pen
- Cello tape
- Screwdriver